

Some Applications of Trigonometry

Using trigonometry to find heights and distances, angles of elevation and depression, line of sight and the horizontal, with worked examples and practice questions.

Introduction

In the previous chapter we defined the trigonometric ratios for acute angles in a right-angled triangle. In this chapter we use those ratios in real life: to find the height of a building, the distance across a river, or the height a kite is flying at, all without actually measuring them directly. The idea is simple: if you know one distance and one angle in a right-angled triangle, you can find any other side using sin, cos or tan.

The problems in this chapter are all about **heights and distances**. The height is a vertical distance (like the height of a tower or a tree), and the distance is a horizontal distance (like the distance from your feet to the foot of the tower). Together they form the two legs of a right-angled triangle, and the line joining your eye to the top of the object is the hypotenuse.

Key Terms: Line of Sight, Horizontal and Angles

Before solving problems, it helps to be clear about three terms that appear in every question of this chapter. NCERT uses them with a very specific meaning, so you should use them the same way in your answers.

Line of Sight

The straight line from the observer's eye to the object being looked at. If you are looking at the top of a tower, the line of sight is the line joining your eye to the top of the tower.

Horizontal Line

The straight line through the observer's eye that is parallel to the ground. Think of it as the line you see when you look straight ahead, not up or down.

Angle of Elevation and Angle of Depression

Angle of elevation: the angle between the horizontal line and the line of sight when the object is above the horizontal line. You have to look up, that upward angle is the angle of elevation.

Angle of depression: the angle between the horizontal line and the line of sight when the object is below the horizontal line. You have to look down, that downward angle is the angle of depression.

Both angles are always measured from the horizontal, not from the vertical.

A useful picture to keep in mind:

- If you stand on the ground and look up at a bird, the angle your line of sight makes above the horizontal is the **angle of elevation** of the bird from your eyes.
- If you stand on top of a building and look down at a car on the road, the angle your line of sight makes below the horizontal is the **angle of depression** of the car from your eyes.

One result that comes up in almost every problem involving the angle of depression: **the angle of depression at the top equals the angle of elevation at the bottom**, because the horizontal line at the top and the ground are parallel, and the line of sight is a transversal. So the two alternate interior angles are equal. Drawing this on your diagram and marking the equal angles often makes the right-angled triangle easy to spot.

Also remember: the observer's height is sometimes ignored (the eye is taken to be at ground level) and sometimes given. Read the question carefully. If it says "the observer's eye is 1.5 m above the ground" or "a 1.2 m tall girl", you must add or subtract that height in your diagram.

How to Solve a Heights and Distances Problem

Every problem in this chapter follows the same pattern. If you make a clear diagram and label what you know, the trigonometry is usually just one or two steps.

Step 1. Draw a diagram. Mark the ground as a horizontal line. Mark the vertical object (tower, building, tree) perpendicular to the ground. Mark the observer's eye and draw the horizontal line through it and the line of sight to the object. Label the known heights, distances and angles.

Step 2. Identify the right-angled triangle. The vertical side, the horizontal side and the line of sight form a right angle at the foot of the vertical object (or its projection). Name the triangle clearly, for example $\triangle ABC$ right-angled at B.

Step 3. Choose the right ratio. Ask yourself: which sides do I know and which do I need? Then pick sin, cos or tan accordingly.

- Need opposite and know adjacent (or vice versa) → use **tan** (most common in this chapter).
- Need hypotenuse and know opposite → use **sin**.
- Need hypotenuse and know adjacent → use **cos**.

Step 4. Substitute standard angles if given. The angles in this chapter are almost always 30° , 45° or 60° (occasionally 0° or 90° does not appear as an angle of elevation/depression in a realistic problem), so you can use exact values like $\tan 30^\circ = 1/\sqrt{3}$, $\tan 45^\circ = 1$ and $\tan 60^\circ = \sqrt{3}$.

Step 5. Solve and check. Write the answer with the correct unit and a short statement, for example: "So, the height of the tower is 20 m." A quick sense-check helps: a 60° angle of elevation should give a height larger than the horizontal distance (since $\tan 60^\circ = \sqrt{3} \approx 1.73$), while a 30° angle should give a height smaller than the horizontal distance.

Worked Examples

Worked Example 1, Angle of elevation

A tower stands vertically on the ground. From a point on the ground 30 m away from the foot of the tower, the angle of elevation of the top of the tower is 30° . Find the height of the tower.

Let AB be the tower of height h metres, with B on the ground. Let C be the observation point, so $BC = 30$ m and $\angle ACB = 30^\circ$. $\triangle ABC$ is right-angled at B.

$$\tan 30^\circ = AB / BC = h / 30$$

$$\text{Since } \tan 30^\circ = 1/\sqrt{3},$$

$$h / 30 = 1/\sqrt{3} \implies h = 30/\sqrt{3} = 10\sqrt{3} \text{ m} \approx 17.32 \text{ m}$$

So the height of the tower is **$10\sqrt{3}$ m** (about 17.32 m).

Worked Example 2, Angle of depression

From the top of a 20 m high building, the angle of depression of a car on the ground is 45° . How far is the car from the foot of the building?

Let AB be the building of height 20 m, with A at the top and B at the foot. Let C be the car. The angle of depression of C from A is 45° . Draw the horizontal line AD through A. Then $\angle DAC = 45^\circ$ (angle of depression). Since AD is parallel to BC (the ground) and AC is a transversal, the alternate interior angle $\angle ACB = \angle DAC = 45^\circ$. So the angle of elevation of A from C is also 45° .

In right-angled $\triangle ABC$, right-angled at B:

$$\tan 45^\circ = AB / BC = 20 / BC$$

$$\text{Since } \tan 45^\circ = 1, \text{ we get } 20 / BC = 1, \text{ so } BC = 20 \text{ m.}$$

The car is **20 m** from the foot of the building.

Worked Example 3, Two angles (the common "two positions" type)

The angles of elevation of the top of a tower from two points at distances 4 m and 9 m from the foot of the tower and on the same straight line with it are complementary. Find the height of the tower.

Let AB be the tower of height h , with B on the ground. Let C and D be the two points on the ground on the same side of the tower such that $BC = 4$ m and $BD = 9$ m. Let $\angle ACB = \theta$, then $\angle ADB = 90^\circ - \theta$ (complementary). $\triangle ABC$ and $\triangle ABD$ are both right-angled at B.

$$\tan \theta = h / 4 \quad \dots(1) \quad \text{and} \quad \tan(90^\circ - \theta) = h / 9$$

But $\tan(90^\circ - \theta) = \cot \theta = 1 / \tan \theta$, so $h / 9 = 1 / \tan \theta$. From (1), $\tan \theta = h/4$, so

$$h / 9 = 4 / h \implies h^2 = 36 \implies h = 6 \text{ m} \quad (h > 0)$$

The height of the tower is **6 m**. A useful result to remember: when two angles of elevation from distances a and b are complementary, the height is $\sqrt{ab}$.

Worked Example 4, Observer's height matters

A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Let AB be the building of height 30 m, with B on the ground. Let the boy's eye be at E, 1.5 m above the ground, so the height from his eye level to the top is $AB - 1.5 = 28.5$ m. Let the horizontal distance from his eye to the building be d . Then in the right-angled triangle formed by his eye, the foot of the perpendicular from his eye to the building, and the top:

At first position (angle 30°): $\tan 30^\circ = 28.5 / d_1$, so $d_1 = 28.5 / \tan 30^\circ = 28.5 \times \sqrt{3}$ m.

At second position (angle 60°): $\tan 60^\circ = 28.5 / d_2$, so $d_2 = 28.5 / \tan 60^\circ = 28.5 / \sqrt{3}$ m.

Distance walked $= d_1 - d_2 = 28.5\sqrt{3} - 28.5/\sqrt{3} = 28.5(\sqrt{3} - 1/\sqrt{3}) = 28.5 \times (2/\sqrt{3}) = 57/\sqrt{3} = 19\sqrt{3}$ m ≈ 32.91 m.

He walked **$19\sqrt{3}$ m** (about 32.9 m) towards the building. If the question ignores the boy's height and takes the eye at ground level, simply use 30 m instead of 28.5 m and the distance walked would be $20\sqrt{3}$ m.

Notice that every example needed only **tan** of a standard angle. That is typical for Chapter 9. In a few problems you may need sin or cos instead, for example when the length of the line of sight (a string, a ladder) is given and you need the vertical height.

Summary

- **Line of sight** is the line from the observer's eye to the object; **horizontal** is the line through the eye parallel to the ground.
- **Angle of elevation** is the angle above the horizontal when the object is above the eye (look up); **angle of depression** is the angle below the horizontal when the object is below the eye (look down). Both are measured from the horizontal.
- The angle of **depression at the top** equals the angle of **elevation at the bottom** (alternate interior angles, since the horizontal at the top is parallel to the ground).

- All heights and distances problems reduce to a **right-angled triangle**: vertical side (height), horizontal side (distance) and line of sight (hypotenuse). Choose sin, cos or tan based on which sides are known and which are needed, tan is the most common.
- Angles given are almost always **30°, 45° or 60°**, so use exact values: $\tan 30^\circ = 1/\sqrt{3}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$, $\sin 30^\circ = 1/2$, $\sin 60^\circ = \sqrt{3}/2$, and so on.
- Read whether the **observer's height** (eye level) is given, if it is, subtract it from the total height before applying trigonometry.
- Always draw a **clear, labelled diagram** with the right angle, the known angle, and the unknown marked. It is worth full marks in the board exam even if the calculation has a small slip.

Practice Questions

1. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming there is no slack.

2. From the top of a 15 m high building, the angle of elevation of the top of a tower is 30° and the angle of depression of the foot of the tower is 45° . Find the height of the tower and its distance from the building.

3. A 1.2 m tall girl looks at a lamp post 10 m away. The angle of elevation of the top of the lamp post from her eyes is 60° . Find the height of the lamp post.

4. From a point on a bridge 20 m above the water, the angles of depression of two boats on opposite sides of the bridge are 30° and 45° . Find the distance between the two boats.

5. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60° . From another point 20 m away from this point on the same bank and in line with the foot of the tower, the angle of elevation is 30° . Find the height of the tower and the width of the canal.

6. The angle of elevation of a cloud from a point 60 m above a lake is 30° , and the angle of depression of its reflection in the lake is 60° . Find the height of the cloud above the lake. (Reflection is as far below the lake surface as the cloud is above it.)